

TALENT: Intelligent Management of Mobility and Parking in Mountain Low-Density Heritage Cities

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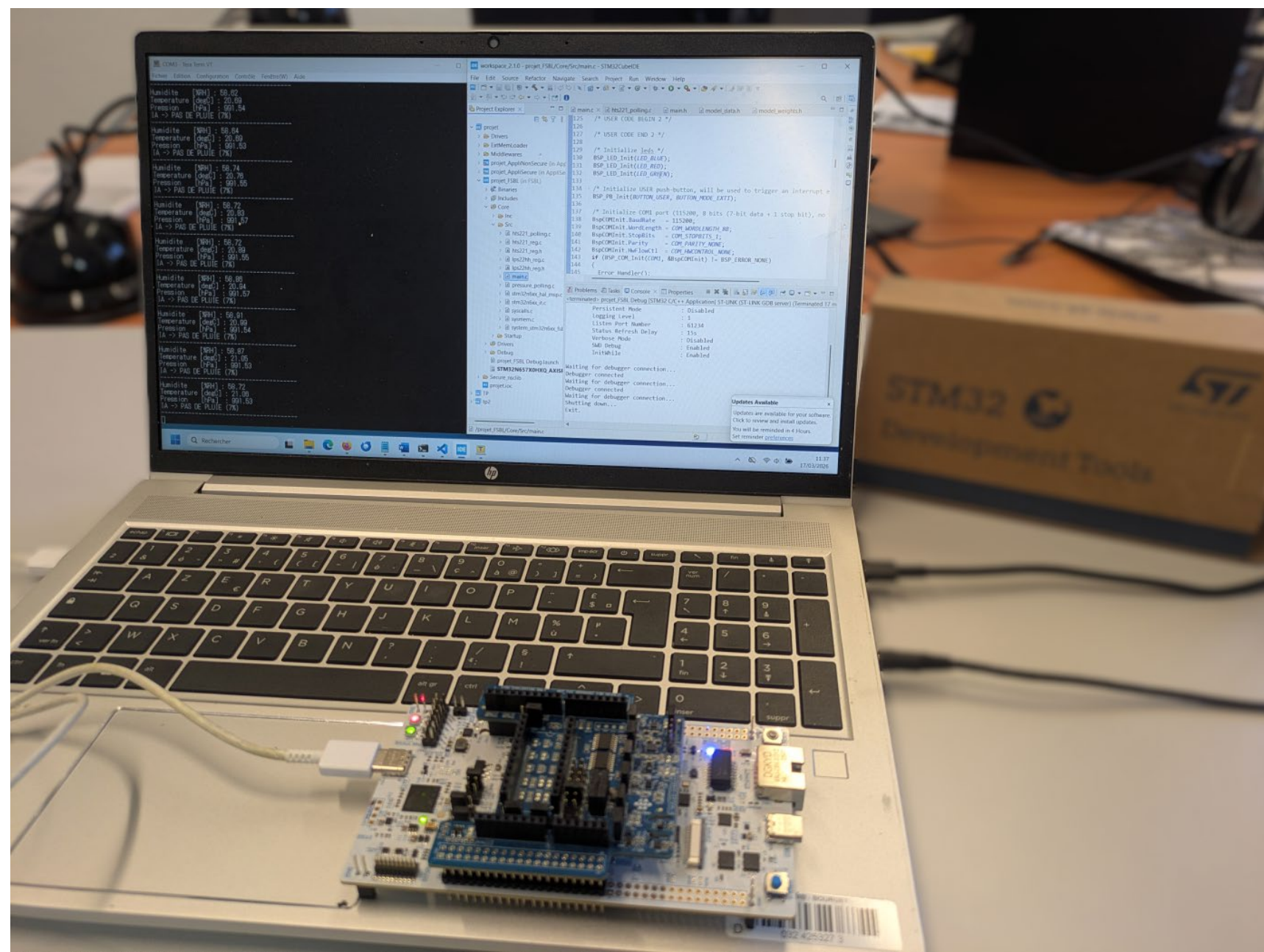
Introduction & Context

Standard weather models lack the local precision needed for mountains, where conditions change very fast. At Mont Revard (1500m), poor network connections make cloud-based forecasting impossible. To solve this, we built a small, smart system that works 100% offline directly on the mountain, providing instant and accurate real-time predictions.



Neural Network Architecture: Weather Prediction Model

The Hardware



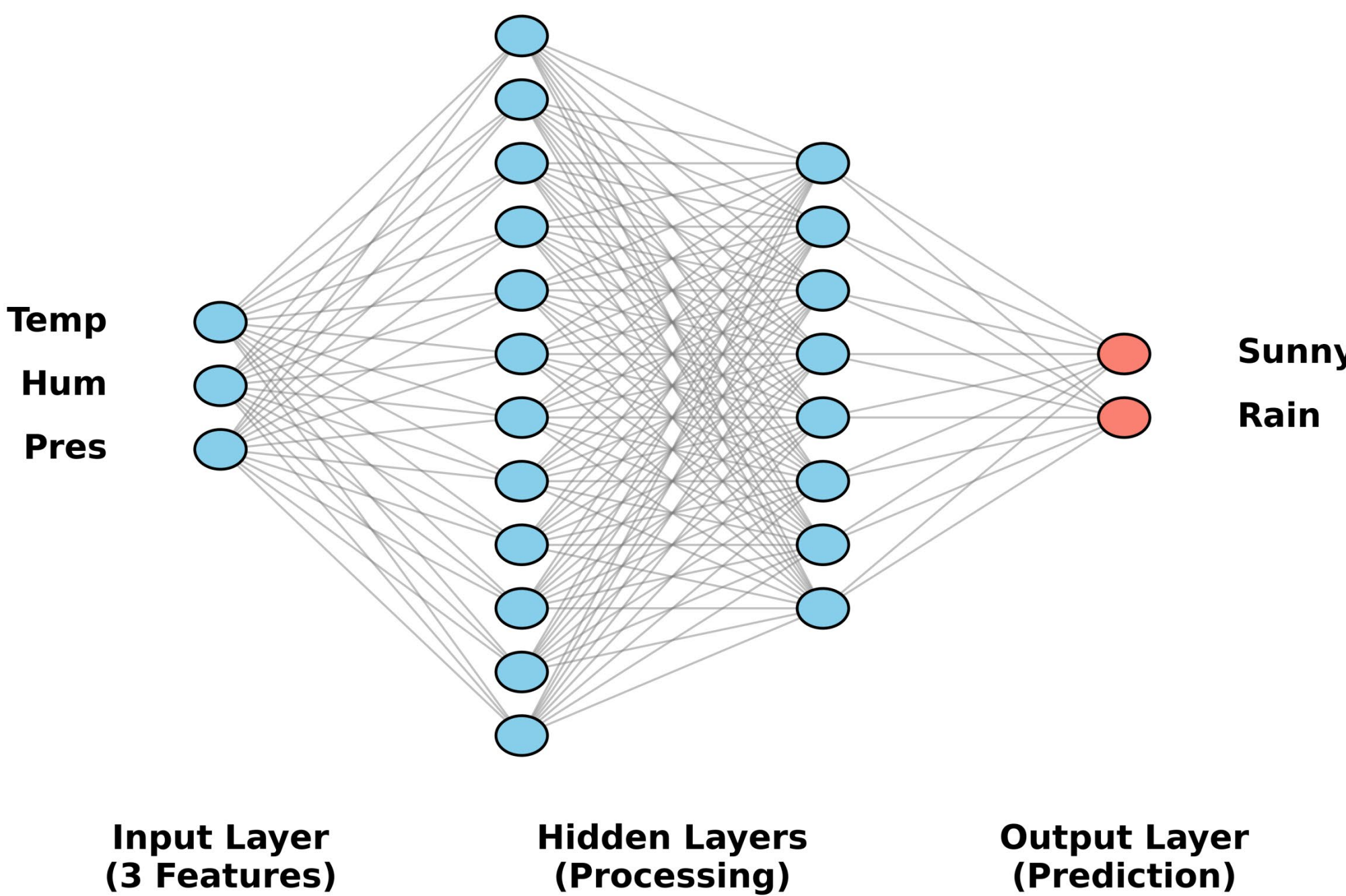
Chip: STM32 N657X0-Q
Boost: ST Neural-ART hardware engine

The first STM32 dedicated to Deep Learning, offering a 600x boost over traditional MCUs.

800 MHz Cortex-M55 Core.

Neural Processing Unit: 600 GOPS throughput.

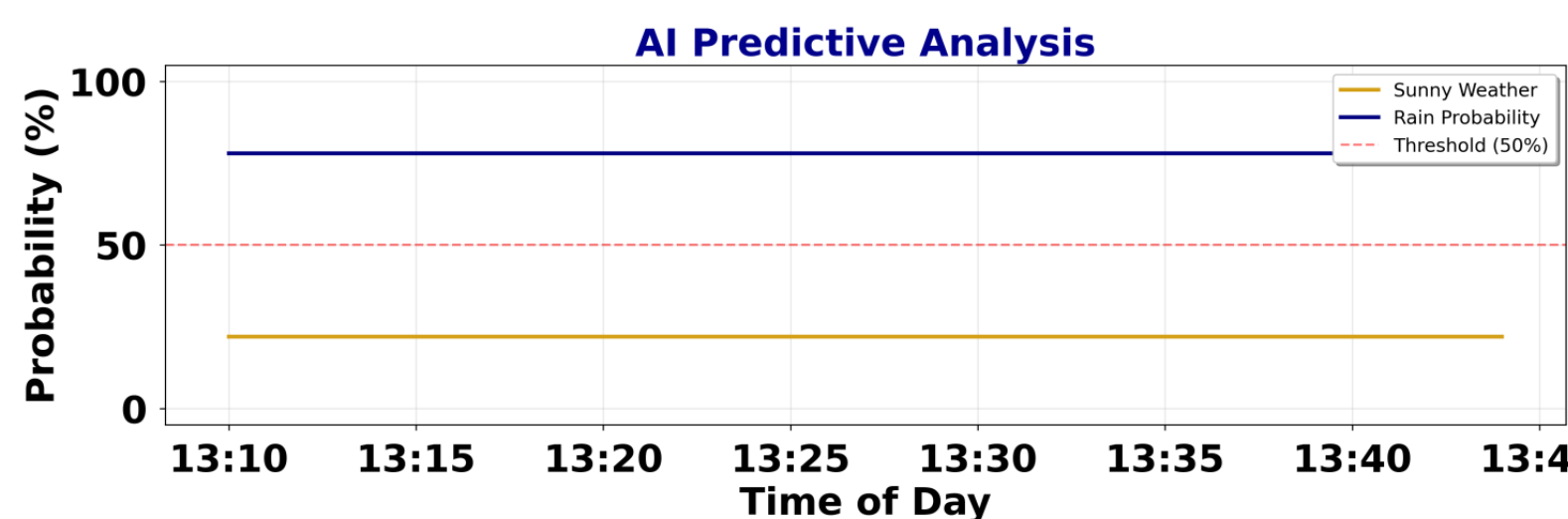
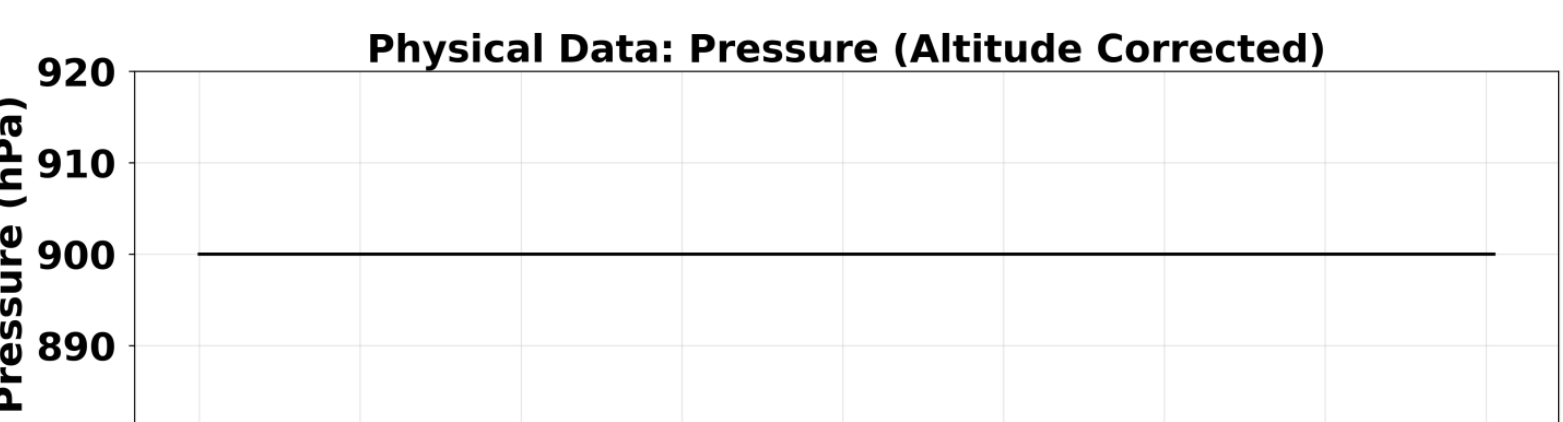
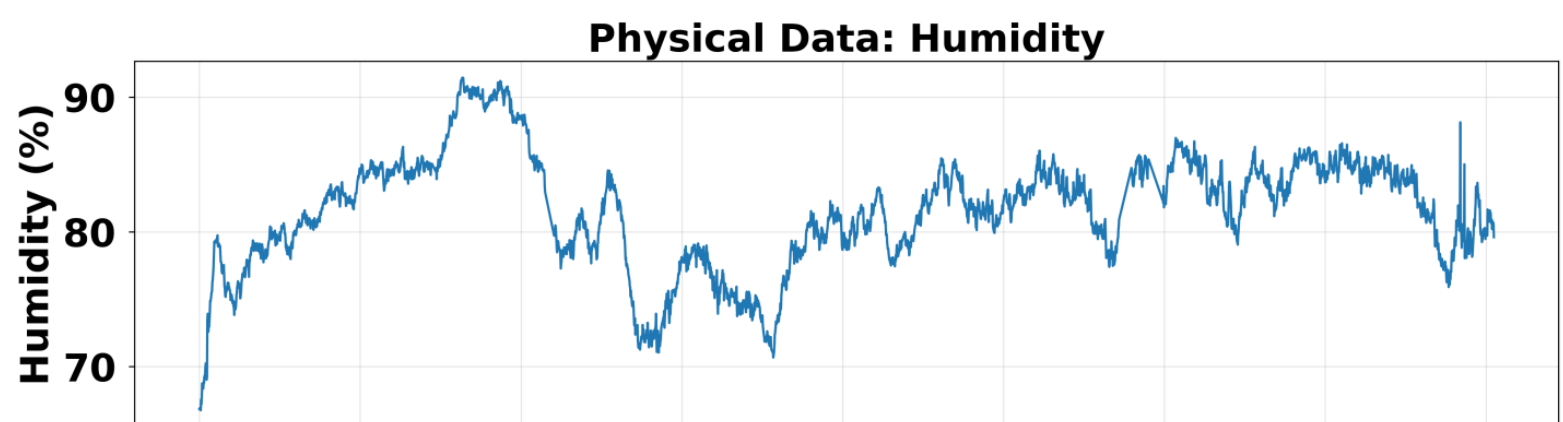
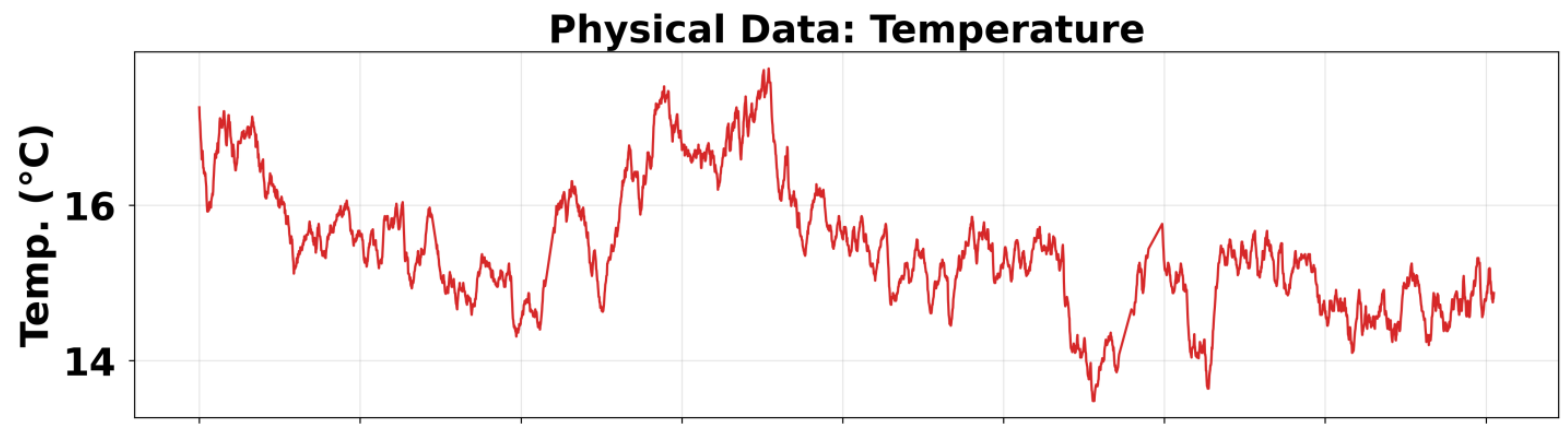
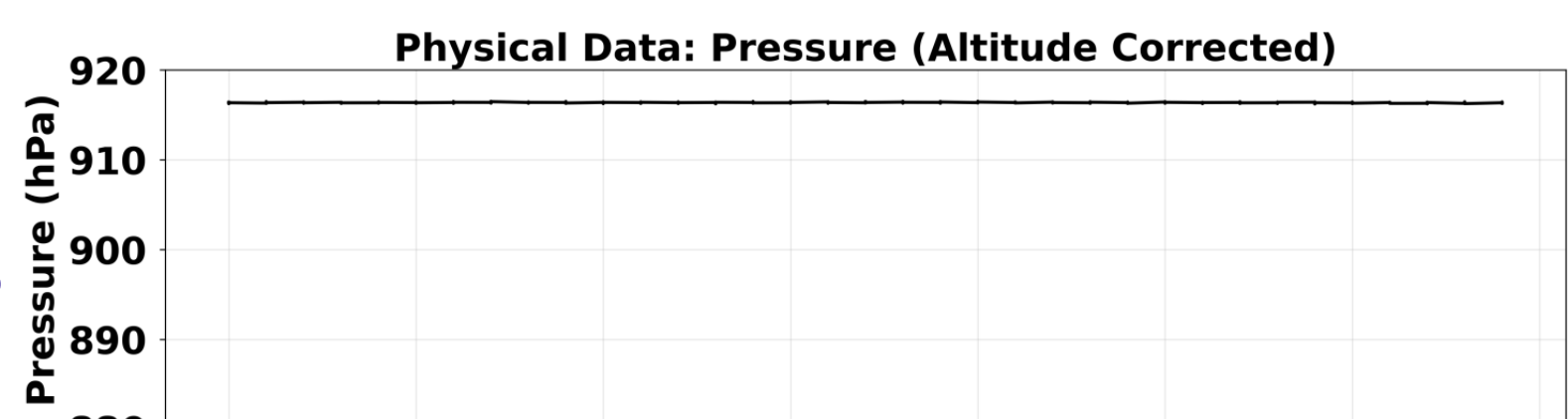
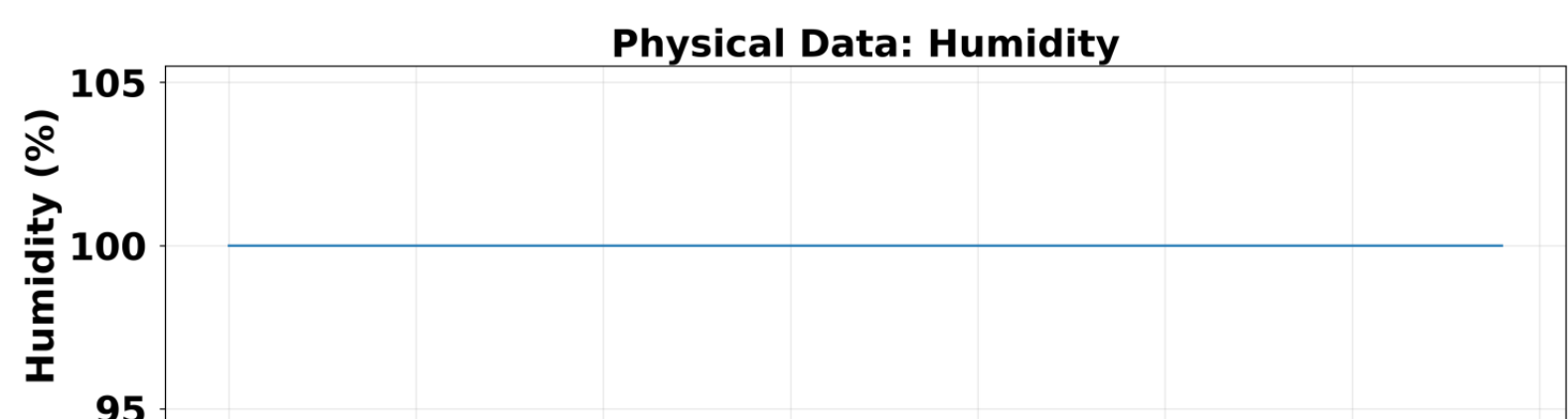
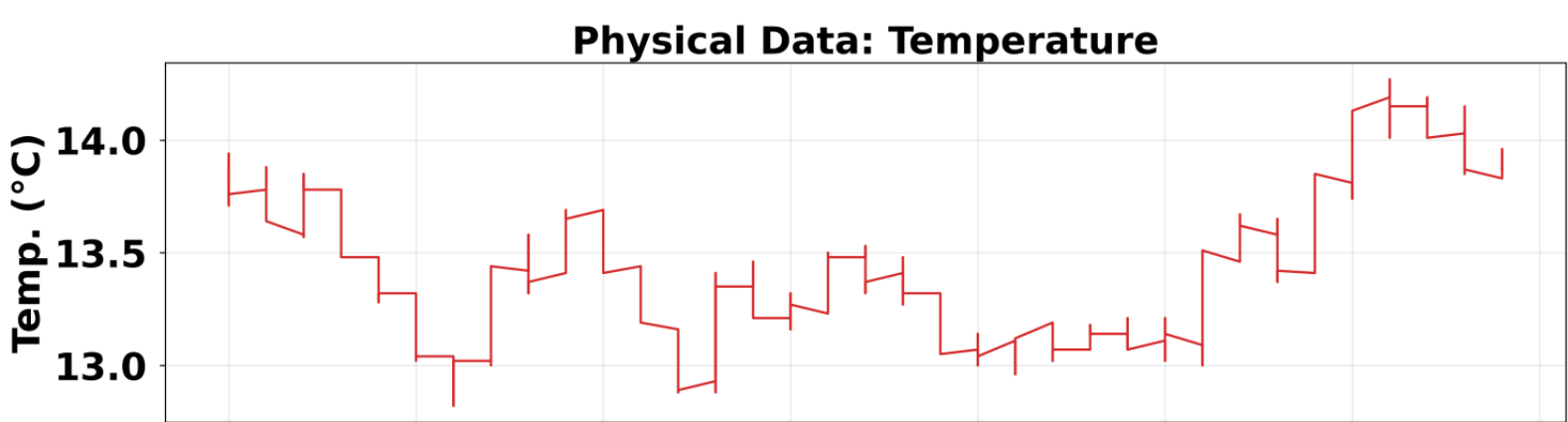
Energy: Optimized for ultra-low power standby modes.



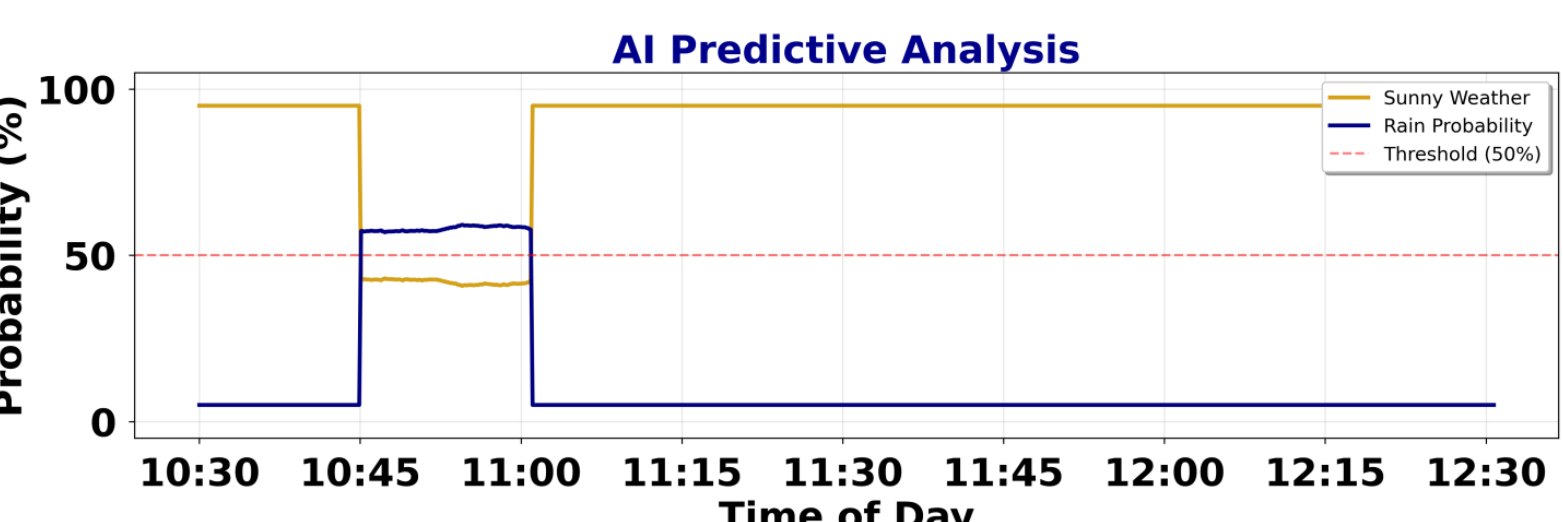
Live Field Tests

1. Rain Transition (10:30 – 12:30):

Observation: Sudden temperature drop and rapid cloud movement.
AI Reaction: Immediate spike in rain probability above the 50% threshold.
Result: High model reactivity to fast, transient weather changes.



Live Field Test: Rain Resilience (13:10 – 13:45)



Live Field Test: Rain Transition (10:30 – 12:30)

2. Fog Resilience (13:10 – 13:45):

Observation: Saturated air with 100% humidity and stable low pressure.
AI Reaction: Model maintains a steady 60% rain prediction.
Result: Filters out sensor noise perfectly to keep a stable decision curve.

Top Accuracy: 99.8%
Very Stable: Learned everything in under 30 steps (epochs).

Detects rain immediately when temperature drops.
Works perfectly even with 100% humidity and bad weather



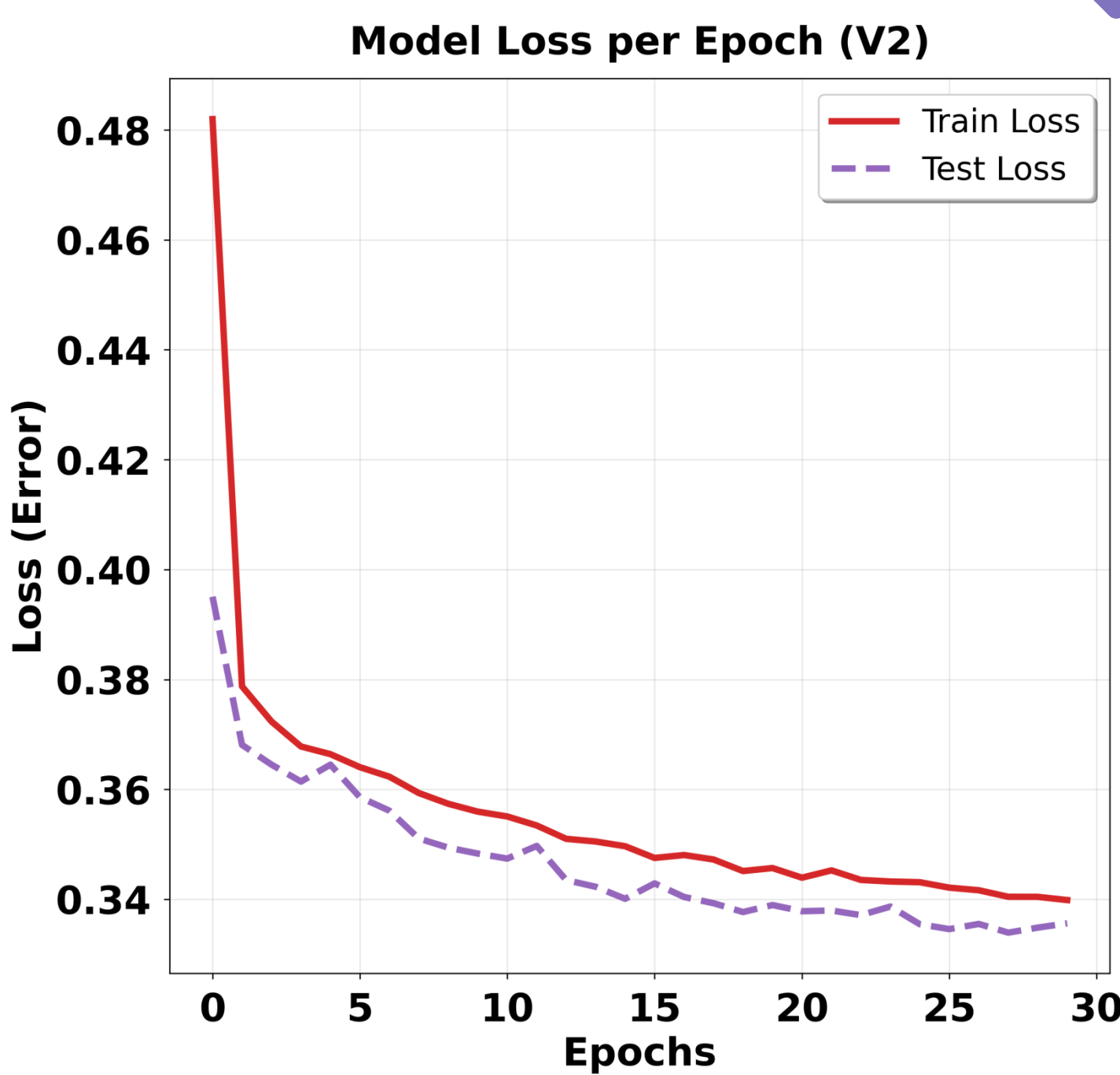
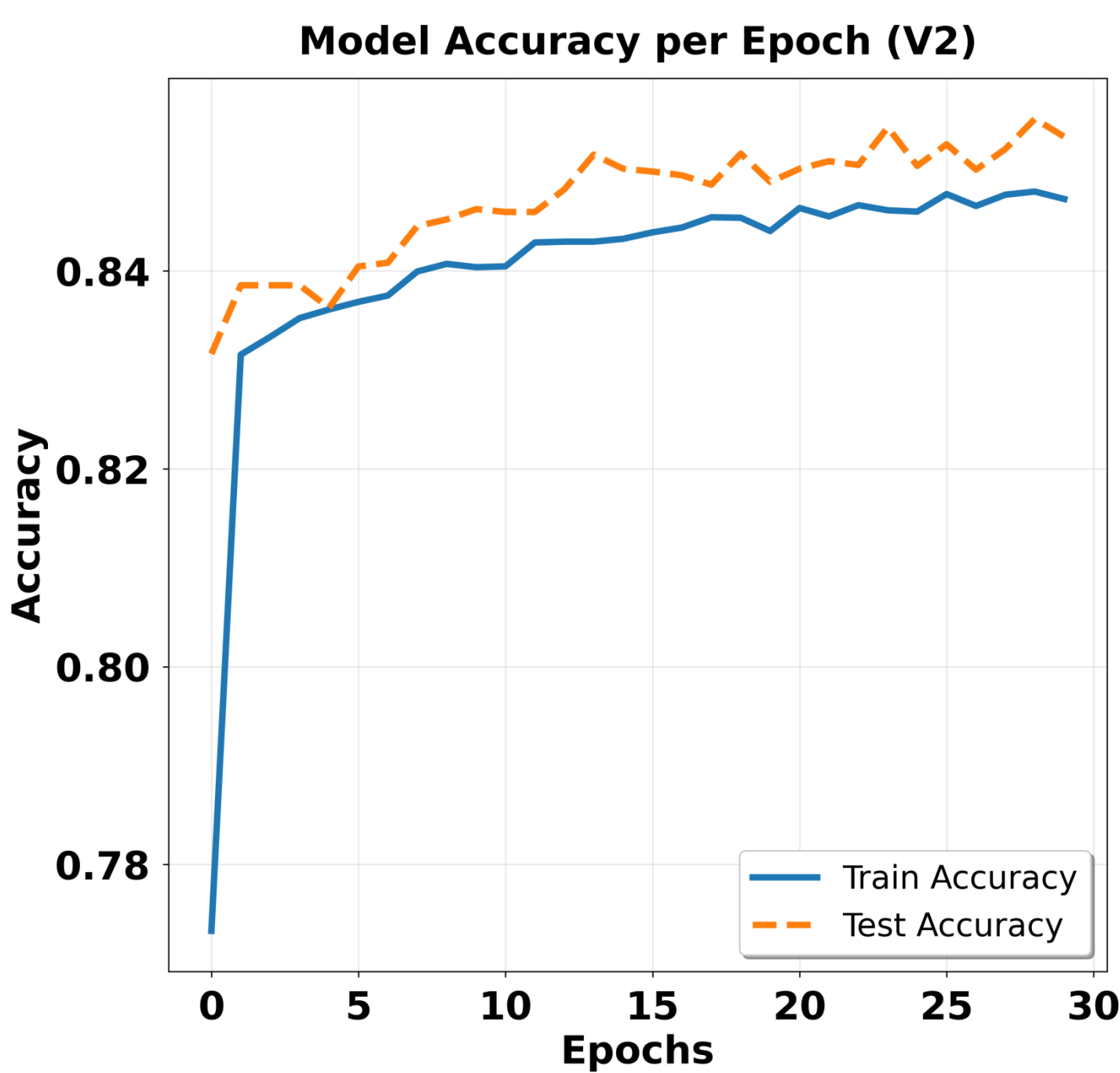
Extreme Fog Conditions (13:10)



Transient Cloud Event (10:45)

Input Layer:
Captures 3 key physical features (Temperature, Humidity, Pressure).
Hidden Layers:
Complex intermediate processing via 2 deep levels (comprising 128 and 64 neurons respectively).
Output Layer:
Binary classification generating the final probability between two distinct states (Sunny / Rain).

Test results



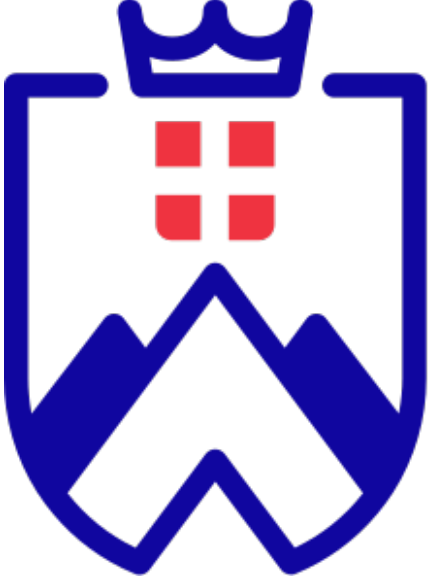
Keras NN V2: Accuracy & Loss Performance Curves



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